Title: NDVI Scale Sensitivity and Edge Effects in Smallholder Agricultural Landscapes: A Sentinel-2 Study from Bangladesh

1. Abstract

This report presents an empirical assessment of Normalized Difference Vegetation Index (NDVI) reliability across varying spatial resolutions in the smallholder agricultural context of Sylhet, Bangladesh. Using Sentinel-2 multi-spectral imagery and NASA Harvest "Fields of the World" (FTW) boundaries, the study evaluates how scale-induced bias and edge contamination affect vegetation monitoring as the mismatch between field size and pixel resolution increases. Analysis of a stratified sample of 200 agricultural fields indicates that NDVI estimates become relatively stable when field areas exceed approximately 180 m^2. Conversely, fields smaller than 90 m^2 and those with elongated geometries exhibit substantially higher bias due to edge effects. Correlation analysis revealed that scale-induced bias is negatively associated with field area ($r = -0.18$, $p < 0.05$) and positively associated with geometric complexity ($r = 0.25$, $p < 0.01$). These findings provide practical constraints for the use of 10m to 30m satellite data in fragmented landscapes.

2. Introduction and Research Question

Monitoring smallholder agriculture using satellite remote sensing remains challenging due to the high degree of landscape fragmentation. While Sentinel-2 provides high-resolution (10m) multi-spectral data, the common practice of spatial aggregation or the use of coarser historical products (e.g., Landsat at 30m) can introduce significant spectral errors. These errors often arise when field dimensions approach the pixel size of the sensor, leading to a "mixed pixel" effect where the signal of the crop is contaminated by adjacent land covers.

The central research question of this study is: **How does NDVI reliability change when field size and spatial resolution mismatch increases in smallholder agricultural landscapes?** Understanding this relationship is critical for agricultural statistics and crop monitoring in regions like Bangladesh, where field sizes are often small and shapes are highly irregular, increasing the likelihood of boundary-induced spectral instability.

3. Study Area and Data Summary

The study was conducted in the Sylhet District of Bangladesh, a region characterized by diverse, small-scale agricultural plots primarily dedicated to rice cultivation.

Table 1: Data Sources and Specifications

Source	Resolution	Details

Sentinel-2 Surface Reflectance	10m, 20m, 30m	Multi-spectral imagery; Date range: 2024-07-01 to 2024-11-30.
NASA Harvest FTW	Vector	AI-generated field boundaries for agricultural plots.
Sylhet District Boundary	Vector	Administrative boundary used for spatial clipping.

Figure 1: Study Area

4. Variable Definitions

The following variables were used to quantify the relationship between field geometry and spectral signal stability.

Table 2: Definition of Analysis Variables

Variable	Definition	Unit
Area	Total surface area of the agricultural field.	m^2
Core Area	Area of the field excluding a 2.5m inward buffer.	m^2
Edge Ratio	The proportion of the field categorized as "edge" relative to total area.	Dimensionless
Shape Index	Measure of geometric complexity and elongation.	Dimensionless
Bias30	The difference between 30m aggregated NDVI and baseline 10m NDVI.	NDVI
Edge Effect	The absolute difference between Core NDVI and Edge NDVI.	NDVI

Mathematical Equation for Shape Index:

The Shape Index was calculated to determine the deviation of a field's geometry from a perfect circle, where a value of 1.0 represents a circular plot and higher values represent increasing elongation or irregularity.

$$\text{Shape Index} = \frac{Perimeter}{2 \times \sqrt{\pi \times Area}}$$

5. Methodology

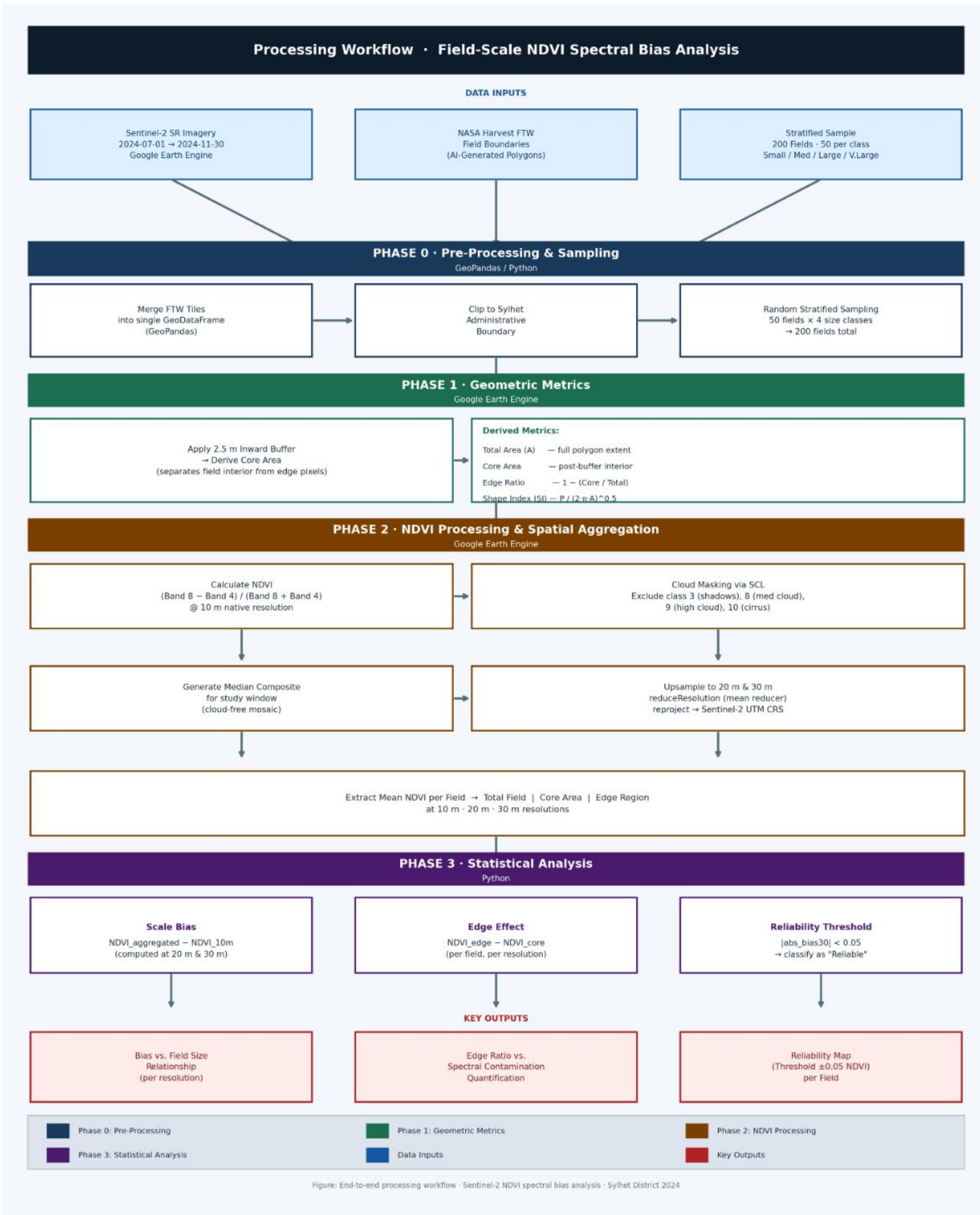

Figure 2: Workflow for NDVI Scale Sensitivity and Edge Effect Analysis

5.1. Phase 0: Data Pre-processing

Field boundaries were obtained as NASA Harvest FTW tiles. These tiles were merged using GeoPandas, and geometry issues (such as self-intersections) were resolved by applying a zero-width buffer (buffer(0)). The dataset was then clipped to the Sylhet District boundary. To ensure a representative analysis, a stratified sample of 200 fields was selected, consisting of 50 fields per the following size classes:

- **Small:** < 300 m^2
- **Medium:** 300 – 1,500 m^2
- **Large:** 1,500 – 5,000 m^2
- **Very Large:** > 5,000 m^2

5.2. Phase 1: Geometric Analysis

Geometric metrics were computed for each sampled field. To distinguish between the interior and the perimeter of the plots, an inward buffer of **2.5m** was applied to define the "Core" area. This distance was chosen to represent a sub-pixel shift at 10m resolution. The "Edge" geometry was then calculated as the geometric difference between the total field boundary and the core polygon. A limitation of this approach is the use of a fixed buffering logic, which may disproportionately reduce the core area of the smallest fields.

5.3. Phase 2: Google Earth Engine (GEE) Processing

Multi-spectral data from the Sentinel-2 Surface Reflectance collection (COPERNICUS/S2_SR) were processed in GEE. Cloud masking was performed using the Scene Classification Layer (SCL) to exclude pixels identified as clouds, shadows, or snow. A median NDVI composite was generated for the study period.

Spatial aggregation was performed to simulate coarser resolutions:

- **10m NDVI:** Baseline resolution.
- **20m/30m NDVI:** Generated using the reduceResolution function with a mean reducer and then reprojected to match the target scales.

Mean NDVI values were extracted for the total, core, and edge geometries using the reduceRegions function at a 10m scale to capture internal variability.

5.4. Phase 3: Statistical Analysis (Python)

Extracted datasets were merged in Python using Pandas. Bias metrics (bias20, bias30) were calculated by subtracting the 10m baseline NDVI from the coarser resolution values. Edge effects were quantified as the difference between core and edge NDVI values. Reliability was assessed using a threshold of abs_bias30 < 0.05. Statistical significance was determined using Pearson correlation and regression analysis.

6. Results

6.1. Field Distribution and Study Area

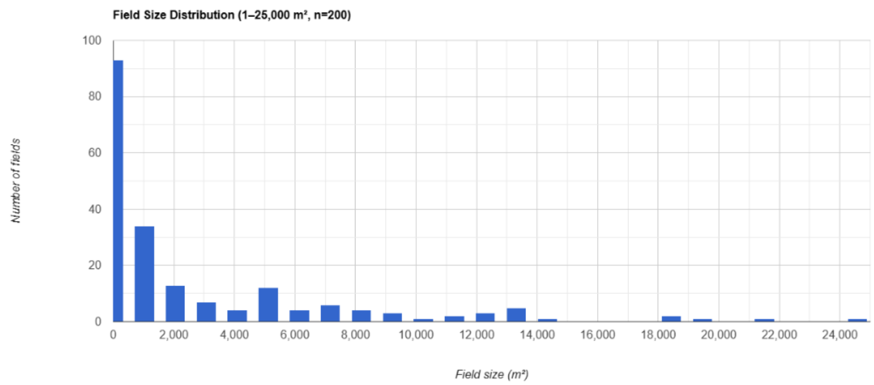

Figure 3: Field Size Distribution

The sampled fields represent a wide range of sizes typical of fragmented landscapes. The distribution is heavily skewed toward smaller area classes, reflecting the subsistence-level land tenure common in the Sylhet region.

6.2. Scale-Induced Bias

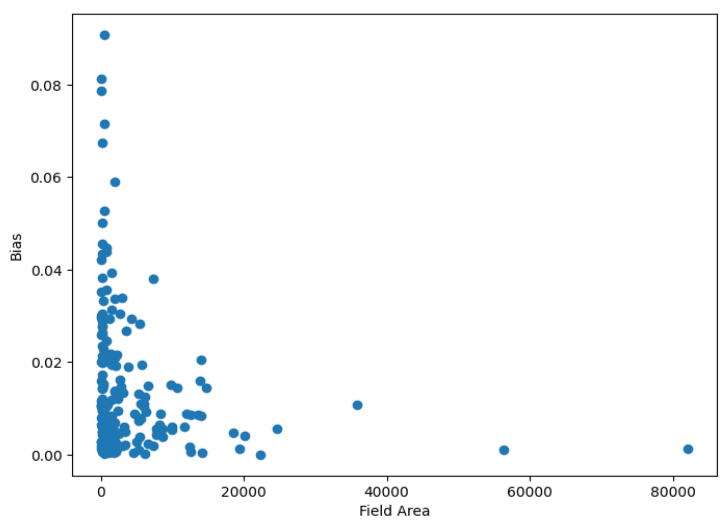

Figure 4. Relationship Between Field Area and Absolute NDVI Bias at 30 m Resolution

As the spatial resolution coarsens from 10m to 30m, NDVI values deviate from the baseline. In our sampled dataset, fields below approximately 90 m² exhibited substantially higher NDVI bias. A negative correlation (r = -0.18) exists between field area and abs_bias30, indicating that as field size decreases, the aggregation error increases.

6.3. Edge Contamination

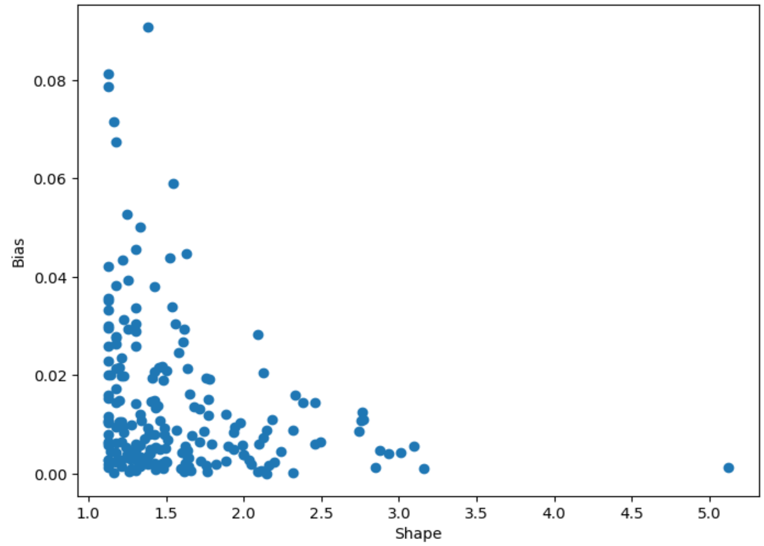

Figure 5: Relationship Between Shape Complexity and Absolute NDVI Bias

Geometric complexity significantly influences signal purity. The "effect of elongated geometries" was observed where fields with high Shape Indices ($r = 0.25$, $p < 0.01$) showed increased edge contamination. This occurs because elongated fields have a higher perimeter-to-area ratio, ensuring that even at 10m resolution, a higher percentage of pixels are "mixed" with boundary features such as paths, water, or neighboring crops.

6.4. Reliability Thresholds

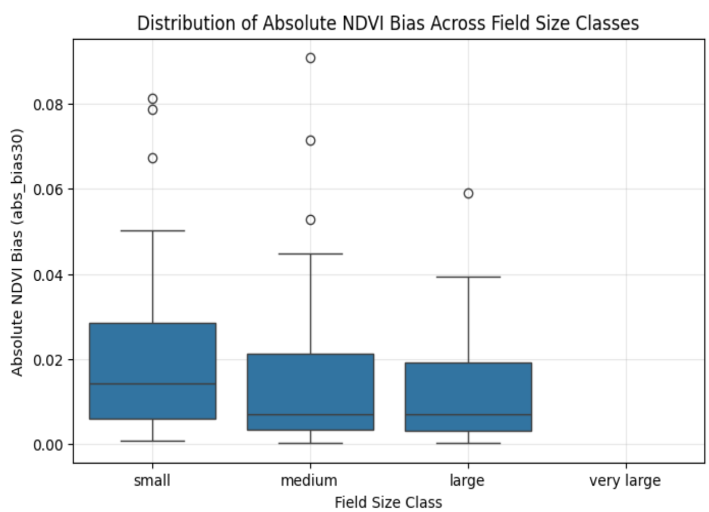

Figure 6: Boxplot by Size Class

Within the analyzed sample, the 90 m^2 threshold was identified as the minimum area where abs_bias30 fell below 0.05. However, consistent reliability was only observed at larger scales; in this study, fields larger than approximately 180 m^2 generally exhibited relatively stable estimates with abs_bias30 < 0.05.

7. Discussion

The results demonstrate that NDVI reliability is highly scale-dependent in smallholder agricultural systems. Irregular field shapes increase edge contamination by increasing the proportion of pixels that intersect with non-crop land cover.

Potential explanations for the observed bias include the sensor's Point Spread Function (PSF), which may lead to signal leakage from adjacent pixels, and geometric registration uncertainties between different satellite acquisitions. Furthermore, as field size decreases relative to the 30m pixel size, the proportion of "mixed pixels" increases mathematically, leading to the instability observed in plots below 90 m^2. The higher edge contamination in elongated fields confirms that perimeter-to-area ratios are just as critical as total area when determining the suitability of medium-resolution data for crop monitoring.

8. Limitations and Future Work

Several limitations affect the generalizability of these findings:

- **Fixed Buffering:** The use of a 2.5-meter fixed buffer targets sub-pixel shifts but may be insufficient for fully isolating the core in the smallest field classes.
- **AI Boundary Uncertainty:** Field boundaries from NASA Harvest FTW are AI-generated and may contain inherent geometric errors or over-segmentation compared to ground-truth data.
- **Temporal and Spatial Scope:** This study is limited to one district (Sylhet) and one growing season.

Future Work: Future research should include validation using manually digitized field boundaries to assess the accuracy of AI-generated inputs. Additionally, sensitivity analyses across different crop types and diverse geographic regions would help determine if the 180 m^2 stability threshold is consistent across different ecological zones.

9. Conclusion

This study provides an empirical assessment of the impact of spatial resolution and field geometry on NDVI reliability in smallholder agricultural landscapes. The data suggests that while Sentinel-2 10m data is effective for monitoring, aggregating to 30m introduces significant bias in fields smaller than 180 m^2. These findings emphasize that agricultural monitoring must account for scale-induced errors in fragmented environments, particularly when the field dimensions are comparable to the sensor's pixel size.

Reproducibility Statement: The workflow described in this report is fully reproducible. The Google Earth Engine scripts, Python notebooks, and data merging workflows are available via the project’s documentation and code repositories.

10. References

- *Drusch, M., et al. (2012). Sentinel-2: ESA's Optical High-Resolution Mission for Data Continuity and Operational Services. Remote Sensing of Environment.*
- *Townshend, J. R. G., & Justice, C. O. (1988). Selecting the Spatial Resolution of Satellite Sensors Required for Global Monitoring of Land Transformations. International Journal of Remote Sensing.*
- *Copernicus Sentinel-2 Surface Reflectance (S2_SR) Data Products.*
- *Google Earth Engine API and Documentation.*
- *NASA Harvest FTW (Fields of the World) Dataset.*
- *Pandas and GeoPandas Python Libraries for Spatial Data Analysis.*